MUSE PORT · EVIDENCE REPORT

Muse ports llm.c from CUDA to SYCL

Meta's `muse-spark-1.3` planned the port of karpathy/llm.c `dev/cuda`, then ported its kernel files one by one in an agent loop. A hidden reference the model never sees grades every port; passing ports were then run on NVIDIA GPUs against llm.c's own CUDA, and on an AMD GPU against the reference.

The plan Every item GPT-2 Differential checks Specs by Muse llm.c-sycl Full training llama2.cu bf16 trainer The first runs Repository

llm.c's production bf16 trainer: ported in one run, tuned on real GPUs

llm.c's `train_gpt2.cu`, `test_gpt2.cu` and `llmc/`: 12 CUDA files and 4,884 lines, in bf16 mixed precision with cuBLASLt, warp reductions and stochastic rounding. The hidden check is llm.c's own test, as Muse ports it, at llm.c's bf16 tolerances: logits, all 16 gradient tensors, 10 losses against PyTorch and a bit-exact checkpoint replay. A guard fails any port that loosens a tolerance. For speed, Muse got a `gpu_test` tool: it builds Muse's current code on a rented GPU and returns llm.c's test result there, llm.c's step time next to llm.c's own CUDA on the same GPU, and an nsys kernel profile. A broker outside the model's process holds the GPU keys and the budget.

PORTABLE SYCL + ONEMATH
1.40-2.90x
llm.c's CUDA time on NVIDIA, no vendor code, and llm.c's test runs training's code: passes on the CPU, L4, AMD Instinct MI300X

PORT, ONE RUN
125 turns
run 5, $5.6727: llm.c's test passes

NVIDIA, FINAL
1.13-1.21x
llm.c's CUDA time, same GPU, 4 GPUs (with cuBLAS interop)

PURE SYCL + ONEMATH
1.36-1.52x
the final port, cuBLAS interop compiled out (our diagnostic), 2 GPUs

AMD MI350X
63.5 ms
per step; run 5: 1274.31 ms

MUSE, ALL OF L1
$50.6455
plan and 7 runs; GPU tuning calls $3.4970

Speed: llm.c's trainer, B=4, T=1024, median ms per step over steps 6-30

PORT	GPU	PORT MS	LLM.C CUDA MS	CUDA'S TIME ×
run 5	RTX A5000	2890.85	80.31	**36.00x**
run 5	AMD Instinct MI350X	1274.31		
portable (pure-oneAPI run, turn 209)	RTX A4000	181.1	116.7	**1.55x**
portable (pure-oneAPI run, turn 209)	AMD Instinct MI300X	190.84		
portable (pure-oneAPI run, turn 209)	AMD Instinct MI350X	63.27		
portable (pure-oneAPI run, turn 209)	H100 80GB HBM3	49.53	17.07	**2.90x**
portable (pure-oneAPI run, turn 209)	L4	199.94	142.39	**1.40x**
portable (pure-oneAPI run, turn 209)	RTX A4000	181.57	117.67	**1.54x**
tuned, no GPU	RTX 2000 Ada	1058.32	187.19	**5.65x**
tuned, no GPU	AMD Instinct MI350X	577.86		
tuned with gpu_test (final)	RTX 4090	49.8	43.5	**1.14x**
tuned with gpu_test (final)	L4	160.1	140.6	**1.14x**
tuned with gpu_test (final)	RTX 2000 Ada	205.5	181.9	**1.13x**
tuned with gpu_test (final)	AMD Instinct MI350X	63.5		
final, again (ours, 3 runs)	RTX A4500	113.33	93.64	**1.21x**
final, cuBLAS interop compiled out (ours)	RTX A4500	139.8	91.8	**1.52x**
final, cuBLAS interop compiled out (ours, 3 runs)	RTX 2000 Ada	250.59	184.14	**1.36x**

Read with: the NVIDIA numbers of the final port call NVIDIA's cuBLAS at runtime through SYCL's backend interop, for two bf16 gemm forms oneMath's released API does not have yet (batched bf16, bf16 output; it has both for fp16, and both are under review upstream in oneMath #785 and #770); pure SYCL + oneMath is

the row below them. Muse uses its fast paths only for large shapes, so llm.c's small test does not run them; forced on, the test still passes (diagnostics below). Rented GPUs vary between calls, so compare ratios; each number is one run. llm.c's CUDA does not build on AMD, so AMD has no ratio.

Runs

RUN	STOP	TURNS	MUSE	WHAT HAPPENED
run1	budget	34	$0.6954	SYCLomatic could not parse the trainer; stopped by the toolkit's 2M-token limit before writing code
run2	max_iterations	200	$11.7852	the whole port written; forward pass right, training NaN at the turn limit
run3	success	65	$1.8552	continued run 2's files: llm.c's test passes
run4	stuck	132	$5.5662	from scratch, with a shell: stopped by the stuck rule mid-debugging
run5	success	125	$5.6727	from scratch, with a shell: llm.c's test passes, in one run
tune	success	192	$10.3893	tuning with no GPU: device memory, own attention kernels
gtune	success	211	$14.5022	tuning with gpu_test: profiled, measured every change on real GPUs

Every gpu_test call of the GPU tuning run

#	VENDOR	LLM.C'S TEST	SPEED GPU	PORT MS	CUDA MS	×	CUBLAS INTEROP	
1	nvidia	PASS	RTX 4000 Ada	572.9	107.4	5.33x		
2	nvidia	FAIL	RTX 4000 Ada	333.7	107.9	3.09x		
3	nvidia	FAIL	RTX A4500	599.5	96.5	6.21x		
4	nvidia	PASS	RTX 2000 Ada	502.0	186.7	2.69x		
5	amd	PASS	AMD Instinct MI350X	592.0				
6	nvidia	PASS	L4	419.8	143.2	2.93x		
7	nvidia	PASS	RTX A4000	258.5	116.5	2.22x		
8	nvidia	PASS	L4	235.1	161.8	1.45x		
9	nvidia	PASS	L4	223.7	143.6	1.56x		
10	nvidia	PASS						
11	nvidia	PASS	L4	200.1	142.4	1.41x		
12	nvidia	PASS	RTX 4090	71.6	44.1	1.62x		
13	nvidia	PASS	RTX 4090	71.8	44.2	1.62x	yes	
14	nvidia	PASS	RTX A4000	178.1	113.7	1.57x	yes	
15	nvidia	PASS	RTX 4000 Ada	144.7	107.4	1.35x	yes	
16	nvidia	PASS					yes	
17	nvidia	PASS	RTX A4000	154.1	116.7	1.32x	yes	
18	amd	compile failed					yes	
19	amd	PASS	AMD Instinct MI350X	63.0			yes	
20	nvidia	PASS	RTX 4090	49.8	43.5	1.14x	yes	final
21	nvidia	PASS	L4	160.1	140.6	1.14x	yes	final
22	nvidia	PASS	RTX 2000 Ada	205.5	181.9	1.13x	yes	final
23	amd	PASS	AMD Instinct MI350X	63.5			yes	final

final: the call built exactly the final port. On AMD the cuBLAS code compiles out. Each call keeps the code it built and the answer Muse saw.

Our diagnostics on the final port

VARIANT (A SMALL EDIT BY US, NOT MUSE)	LLM.C'S TEST	SPEED
cuBLAS interop compiled out (pure SYCL + oneMath)	RTX 2000 Ada: PASS	RTX A4500 139.8 ms 1.52x

fast paths forced on for llm.c's test, NVIDIA	RTX 2000 Ada: PASS	L4 159.9 ms 1.14x
fast paths forced on for llm.c's test, AMD	AMD Instinct MI350X: PASS	AMD Instinct MI350X 63.1 ms

Evidence: evidence/l1-llmc-bf16; the GPU tuning run: gtune.

Results with the fixed loop

TIER	ITEMS	CORRECT, 0 HELPERS	LOOP SUCCESS	NVIDIA PASS	AMD PASS	MUSE COST
Tier 1 Plain kernels	10	10	9	10	10	$1.0064
Tier 2 Warp/sub-group, shared memory	5	4	3	4	4	$1.6857
Tier 3 cuBLAS to oneMath	4	4	3	3	4	$1.0576
All	**19**	**18**	**15**	**17**	**18**	**$3.7497**

Correct, 0 helpers: the hidden grade. The final sources compile, match the reference within llm.c's fp32 tolerance, and no SYCLomatic (dpct) helper is left anywhere in the tree. **Loop success**: the loop's stricter rule, met inside the run on one source hash. **NVIDIA pass**: the port matched both the reference and llm.c's CUDA on a real GPU. **AMD pass**: the port matched the reference on an AMD GPU. One run per item, after the loop fixes (delete_file, ranged read_file, 32k output per turn); the first runs are below.

GPT-2: llm.c's training program

Muse ported llm.c's fp32 GPT-2 training file (`train_gpt2_fp32.cu`) in stages, with the real GPT-2 124M weights and llm.c's debug batch. Each stage is graded against PyTorch's numbers at llm.c's own test tolerance, and each check was first proven to pass llm.c's CPU code.

STAGE	WHAT IS CHECKED	HIDDEN GRADE OF THE FINAL SOURCES	TURNS	MUSE COST	GPU
1 · Forward pass	B=4, T=64 debug batch from gpt2_124M_debug_state.bin; logits and loss checked against PyTorch within 1e-2, as test_gpt2_fp32.cu does	pass PASS: 1/1 case(s) match (worst max abs err 0.00121, atol 0.01)	28	$0.7441	CPU device only
2 · Backward pass	B=4, T=64 debug batch from gpt2_124M_debug_state.bin; gradients of all parameters and the loss checked against PyTorch within 2e-2 (llm.c's CPU test tolerance for gradients), as test_gpt2_fp32.cu does	pass PASS: 1/1 case(s) match (worst max abs err 0.0124, atol 0.02)	27	$1.2134	CPU device only
3 · Training: 10 AdamW steps	B=4, T=64 debug batch from gpt2_124M_debug_state.bin; the loss of each of 10 AdamW steps checked against PyTorch's within 1e-2, as test_gpt2_fp32.cu does	pass PASS: 1/1 case(s) match (worst max abs err 0.000154, atol 0.01)	29	$1.1489	pass MI350X VF vs PyTorch pass RTX 4000 Ada vs PyTorch and llm.c's CUDA

Differential checks: the original CUDA as the reference

The hidden references above are C++ programs written from llm.c's CPU code. A differential check needs none: the original CUDA is built with `nvcc` on an NVIDIA GPU and its outputs are recorded, run twice to show they repeat. The recording then grades the port with the same steps as the hidden grade.

Evidence: evidence/diff-llmc-tier1, ports from `evidence/port-llmc-tier1-run2`.

ITEM	RECORDED ON	TWO RUNS IDENTICAL	RECORDING VS CPU REFERENCE	MUSE'S PORT VS RECORDING
`t1-crossentropy-forward`	RTX A4500	yes	pass worst 1.19e-07	pass worst 1.19e-07
`t1-crossentropy-softmax-backward`	RTX 2000 Ada	yes	pass worst 0	pass worst 0
`t1-permute`	RTX A4500	yes	pass worst 0	pass worst 0

t1-residual-forward	RTX A4500	yes	pass worst 0	pass worst 0
t1-gelu-forward	RTX A4500	yes	pass worst 1.19e-07	pass worst 1.19e-07
t1-gelu-backward	RTX 2000 Ada	yes	pass worst 2.38e-07	pass worst 2.38e-07
t1-encoder-forward	RTX A4500	yes	pass worst 0	pass worst 0
t1-encoder-backward	RTX 2000 Ada	yes	pass worst 0	pass worst 4.77e-07
t1-adamw	RTX A4500	yes	pass worst 5.96e-08	pass worst 1.79e-07
t1-fused_residual-forward	RTX 2000 Ada	yes	pass worst 1.43e-06	pass worst 1.43e-06

Planted bugs

BUG	VS CPU REFERENCE	VS RECORDING
subtle: 0.044715f -> 0.0447f	pass worst 3.22e-06	pass worst 3.22e-06
dropped digit: 0.044715f -> 0.04715f	fail 1,062,632 of 1,572,864 values out of tolerance	fail 1,062,644 of 1,572,864 values out of tolerance

A bug whose error is below the tolerance passes both references: that is the limit of any tolerance-based check.

Hidden checks written by Muse

Every hidden check is built from a short spec per file: the CPU reference functions, the kernel variants, the inputs and sizes, the two calls and the tolerance. Muse wrote these specs from the CUDA file alone (two worked examples, not graded). A spec counts only once proven: llm.c's GPU code and CPU code must agree under it, through SYCLomatic on the CPU, or on an NVIDIA GPU where SYCLomatic fails. **As strict as llm.c**: the same tolerance and every fp32 variant, compared with the hand-written spec.

PROVEN	AS STRICT AS LLM.C
16 of 17	16 of 16

FILE	SPEC	VARIANTS	AS STRICT AS LLM.C	ATTEMPTS	MUSE COST	RUN
gelu_backward.cu	proven cpu	1, 2	yes	1	$0.0229	spec-llmc-fp32
residual_forward.cu	proven cpu	1, 2	yes	1	$0.0240	spec-llmc-fp32
permute.cu	proven cpu	1	yes	1	$0.0220	spec-llmc
crossentropy_forward.cu	proven cpu	1	yes	1	$0.0215	spec-llmc
crossentropy_softmax_backward.cu	proven cpu	1	yes	1	$0.0264	spec-llmc
encoder_forward.cu	proven cpu	1, 2, 3	yes	1	$0.0309	spec-llmc
encoder_backward.cu	proven cpu	1, 2	yes	1	$0.0204	spec-llmc
adamw.cu	proven cpu	1, 2	yes	1	$0.0413	spec-llmc
fused_residual_forward.cu	proven cpu	1, 2, 3, 4, 5, 6	yes	1	$0.0399	spec-llmc
layernorm_forward.cu	proven cpu	1, 2, 3, 4, 5, 6	yes	1	$0.0283	spec-llmc
softmax_forward.cu	proven gpu, after the run (first spec)	1, 2, 3, 4, 5, 6, 7, 8	yes	3	$0.1289	spec-llmc
classifier_fused.cu	proven gpu, after the run (first spec)	1, 2, 3, 4, 5	yes	3	$0.1941	spec-llmc
layernorm_backward.cu	proven cpu	1, 2, 3, 5, 6, 7, 8, 9, 10	yes	3	$0.2026	spec-llmc
matmul_forward.cu	proven gpu	1, 2, 3, 4	yes	1	$0.0320	spec-llmc
matmul_backward.cu	proven cpu	1	yes	1	$0.0352	spec-llmc

`trimat_forward.cu`	not proven			3	$0.1992	spec-llmc-fp32
`matmul_backward_bias.cu`	proven gpu	1, 2, 3, 4, 5, 7, 8, 9	yes	1	$0.0553	spec-llmc-fp32

Each file's latest run is shown; spec-llmc-fp32 reran four files with the fp32 build stated in the prompt. The cost column is that run's.

llm.c-sycl: the ports put back together

llm.c-sycl is llm.c laid out as llm.c is, in SYCL, built with `make` and `icpx -fsycl`; its programs run on a CPU, an NVIDIA GPU and an AMD GPU. It holds the kernel ports in `dev/sycl`, and two programs that Muse wrote on top of its proven GPT-2 port: llm.c's fp32 trainer, with its real `main`, and llm.c's own test, which checks logits, loss and every gradient against PyTorch, then 10 training steps.

DEV/SYCL KERNELS	TRAINER AND TEST	MUSE
19 of 19	**3 targets each**	**$2.3292**
pass their hidden checks, 0 helpers	pass: CPU, NVIDIA, AMD	4 loop runs

PROGRAM	CPU (HIDDEN GRADE)	NVIDIA	AMD	MUSE RUNS	EVIDENCE
llm.c's fp32 trainer `train_gpt2fp32sycl`	pass worst 0.00227	pass worst 0.0023 RTX 2000 Ada vs llm.c's CUDA: pass worst 0.00147	pass worst 0.00242 MI350X VF	run1: success 19 turns, $0.7065	evidence/j3-trainer
llm.c's own test `test_gpt2fp32sycl`	pass worst 0.000154	pass worst 0.000127 RTX 2000 Ada vs llm.c's CUDA: pass worst 7.03e-06	pass worst 2.36e-05 MI350X VF	run1: no success 20 turns, $0.4936 run2: no success 30 turns, $0.7003 run3: success 24 turns, $0.4287	evidence/j3-test

Hidden grade: the trainer's validation and training losses against llm.c's own CPU trainer; the test's `overall okay` and its 10 losses against PyTorch's; both at llm.c's tolerance, 1e-2. The GPU runs build the sources Muse last compiled, unchanged, with the toolkit's `gpu-test`; llm.c-sycl's own Makefile also builds and passes the test on both GPUs (evidence/j3-makefile-gpu). llm.c's CUDA does not hipify into something that builds, so on AMD there is no comparison with it.

From a fresh clone (CPU device): `./dev/download_starter_pack.sh` (10 s), `make` (26 s), `./test_gpt2fp32sycl` (34 s): `overall okay: 1` (evidence/j3-fresh-clone).

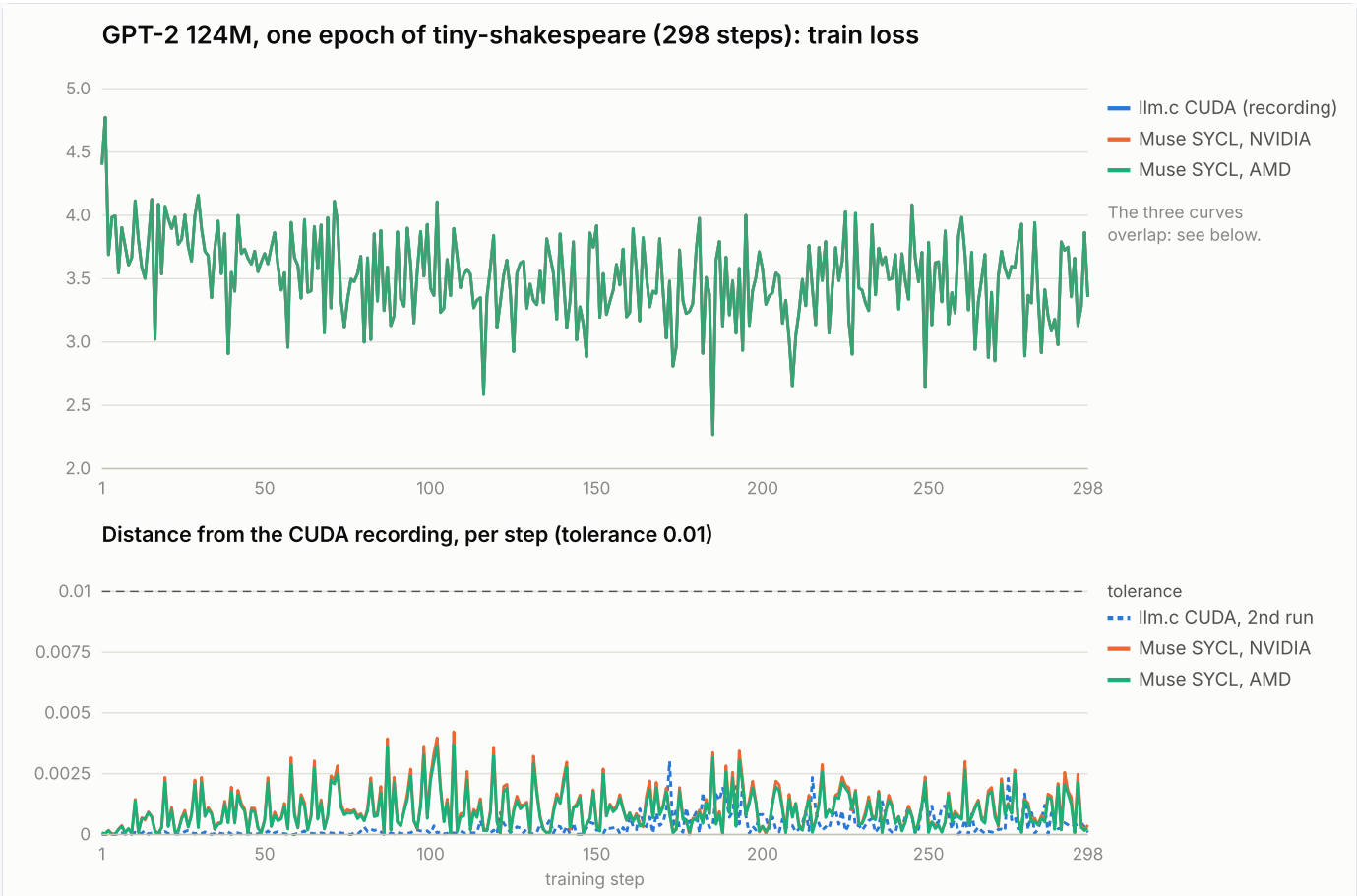
Each check, proven first

PROGRAM	CANDIDATE	RESULT
llm.c's fp32 trainer	llm.c's train_gpt2.c, same 10 steps (known good)	pass worst max abs err 0
llm.c's fp32 trainer	planted bug: gpt2_update learning rate 3e-4 instead of 1e-4	fail 10 of 12 values outside atol 1e-2
llm.c's fp32 trainer	planted bug: train DataLoader without shuffling	fail 11 of 12 values outside atol 1e-2
llm.c's own test	llm.c test_gpt2.c on train_gpt2.c (CPU)	pass overall okay 1, 0 of 10 losses out of tolerance
llm.c's own test	planted bug: GELU coefficient 0.04715 instead of 0.044715, forward and backward (planted/gelu_bug.diff)	fail overall okay 0, 0 of 10 losses out of tolerance

llm.c's own CPU programs must pass and planted bugs must fail before a check grades Muse. A bug can keep every loss within 1e-2 yet fail llm.c's test, so the test check also needs `overall okay: 1`.

Full training: one epoch on NVIDIA and AMD

GPT-2 124M trains for one epoch of tiny-shakespeare (298 steps). llm.c's own CUDA trainer was recorded on an NVIDIA GPU; Muse's SYCL trainer from llm.c-sycl, unchanged, then ran on an NVIDIA and an AMD GPU and was compared with that recording, every val and train loss, at llm.c's tolerance (1e-2).



RUN	GPU	WORST VS RECORDING	PER STEP
llm.c CUDA, second run (noise floor)	RTX 2000 Ada	0.0030	111 ms
llm.c CUDA	RTX A5000	0.0014	54 ms
Muse SYCL	RTX A5000	0.0042	78 ms
Muse SYCL	MI350X VF	0.0037	27 ms

Times are the median of the program's own step timer, one run each: compare them only on the same GPU (the RTX A5000 ran both programs). Evidence: evidence/k1-curves.

A second CUDA repo (one 772-line file): llama2.cu

ankan-ban/llama2.cu is Llama 2 inference in one CUDA file; the rest of the repo is llama2.c's C program and Python scripts, not CUDA. Muse Port had never run on it. Muse planned it as 1 item, grounded against the scan, for $0.0286. The hidden check runs greedy decoding with the stories15M model on two prompts, 64 steps each, and compares the story byte for byte with llama2.cu's own output, recorded on an NVIDIA GPU. llama2-sycl is llama2.cu's repo with the port in place of `llama2.cu`.

CUDA IN THE REPO	MUSE'S PORT	IDENTICAL STORIES
1 file	737 lines	3 devices
llama2.cu: 772 lines, 8 kernels, fp16, CUB	20 turns, one run, 0 helpers	CPU, NVIDIA, AMD; byte for byte

MUSE
$0.4786
plan and port

PORT	CPU (HIDDEN GRADE)	NVIDIA	AMD
`llama2.cpp` run1: success 20 turns, $0.4500	pass worst 0	pass worst 0 RTX 2000 Ada vs llama2.cu: pass worst 0	pass worst 0 MI350X VF

The GPU runs build the sources Muse last compiled, unchanged. On NVIDIA the port is also compared with llama2.cu itself, built with nvcc on the same GPU. llama2.cu's automatic HIP conversion did not build, so on AMD there is no such comparison. Evidence: evidence/j4-llama2.

From a fresh clone of this repository, llama2-sycl's Quick start (`make llama2`, then the two prompts) on the CPU: `both stories identical` (evidence/j4-fresh-clone).

The check, proven first

CANDIDATE	RESULT
llama2.c's fp32 run.c (known good)	`pass` identical, identical
planted bug: RMSNorm epsilon 1e-2 instead of 1e-5	`fail` differs at byte 16, differs at byte 33
planted bug: attention scores divided by sqrt(2*head_size)	`fail` differs at byte 87, differs at byte 53

llama2.c's fp32 `run.c`, a known-good program, must produce the same stories, and planted one-line bugs must not. At 128 steps, `run.c` differs from llama2.cu's fp16 output on a near-tie token, so the check uses 64 steps; that was set before any model ran.

Every item

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU
t1-crossentropy-forward crossentropy_forward.cu	`correct, 0 helpers`	`1 variant kept`	`success`	9	$0.0904	`pass: MI350X VF (vs re` `pass: RTX 2000 Ada`
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	`correct, 0 helpers`	`1 variant kept`	`success`	6	$0.0654	`pass: MI350X VF (vs re` `pass: RTX 2000 Ada`
t1-permute permute.cu	`correct, 0 helpers`	`1 variant kept`	`success`	7	$0.0526	`pass: MI350X VF (vs re` `pass: RTX A4500`
t1-residual-forward residual_forward.cu	`correct, 0 helpers`	`2 variants kept`	`success`	9	$0.0887	`pass: MI350X VF (vs re` `pass: RTX A4500`
t1-gelu-forward gelu_forward.cu	`correct, 0 helpers`	`2 variants kept`	`success`	7	$0.0899	`pass: MI350X VF (vs re` `pass: RTX A4500`
t1-gelu-backward gelu_backward.cu	`correct, 0 helpers`	`2 variants kept`	`success`	11	$0.1502	`pass: MI350X VF (vs re` `pass: RTX A4500`
t1-encoder-forward encoder_forward.cu	`correct, 0 helpers`	`3 variants kept`	`success`	7	$0.0842	`pass: MI350X VF (vs re` `pass: RTX A4500`
t1-encoder-backward encoder_backward.cu	`correct, 0 helpers`	`2 variants kept`	`success`	7	$0.0782	`pass: MI350X VF (vs re` `pass: RTX 2000 Ada`
t1-adamw adamw.cu	`correct, 0 helpers`	`2 variants kept`	`success`	7	$0.0832	`pass: MI350X VF (vs re` `pass: RTX 2000 Ada`
t1-fused-residual-forward fused_residual_forward.cu	`correct, 0 helpers`	`6 variants kept`	`max_iterations`	12	$0.2235	`pass: MI350X VF (vs re` `pass: RTX A4500`
t2-global-norm global_norm.cu	`correct, 0 helpers`	`4 variants kept`	`max_iterations`	16	$0.3230	`pass: MI350X VF (vs re` `pass: RTX 2000 Ada`
t2-layernorm-forward layernorm_forward.cu	`correct, 0 helpers`	`6 variants kept`	`success`	7	$0.1524	`pass: MI350X VF (vs re` `pass: RTX A4500`
t2-softmax-forward softmax_forward.cu	`verify fail`	`6 bodies for 8 variants`	`max_iterations`	16	$0.5857	`not run`
t2-classifier-fused classifier_fused.cu	`correct, 0 helpers`	`5 variants kept`	`success`	7	$0.1397	`pass: MI350X VF (vs re` `pass: GeForce RTX 4090`
t2-layernorm-backward layernorm_backward.cu	`correct, 0 helpers`	`8 bodies for 9 variants`	`success`	14	$0.4848	`pass: MI350X VF (vs re` `pass: L4`

t3-matmul-backward matmul_backward.cu	correct, 0 helpers	1 variant kept	success	8	$0.1108	pass: MI350X VF (vs re fail: RTX A4500
t3-trimat-forward trimat_forward.cu	correct, 0 helpers	5 variants kept	max_iterations	16	$0.3740	pass: MI350X VF (vs re pass: RTX A4500
t3-matmul-backward-bias matmul_backward_bias.cu	correct, 0 helpers	8 variants kept	success	13	$0.3278	pass: MI350X VF (vs re pass: RTX 2000 Ada
t3-matmul-forward matmul_forward.cu	correct, 0 helpers	4 variants kept	success	12	$0.2449	pass: MI350X VF (vs re pass: RTX 2000 Ada

The first runs

The same tasks and hidden checks, before the loop fixes. Most ports were already correct, but the runs spent their turns clearing SYCLomatic's output file by file and hit the toolkit's token limit.

TIER	ITEMS	CORRECT, 0 HELPERS	LOOP SUCCESS	NVIDIA PASS	MUSE COST
Tier 1 Plain kernels	10	9	4	9	$1.6698
Tier 2 Warp/sub-group, shared memory	5	4	2	4	$2.4454
Tier 3 cuBLAS to oneMath	4	4	2	1	$1.3960
All	**19**	**17**	**8**	**14**	**$5.5112**

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	max_iterations	12	$0.1315	pass: RTX 2000 Ada
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	success	12	$0.1916	pass: RTX 2000 Ada
t1-permute permute.cu	correct, 0 helpers	1 variant kept	success	10	$0.1188	pass: RTX 2000 Ada
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	max_iterations	12	$0.1747	pass: RTX A4500
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	success	9	$0.1559	pass: RTX A4500
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	max_iterations	12	$0.1293	pass: RTX 2000 Ada
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	max_iterations	12	$0.1710	pass: RTX 2000 Ada
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	success	12	$0.1210	pass: RTX 2000 Ada
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	max_iterations	12	$0.1266	pass: RTX 2000 Ada
t1-fused-residual-forward fused_residual_forward.cu	correct, 53 helpers	6 variants kept	max_iterations	12	$0.3494	not run
t2-global-norm global_norm.cu	correct, 0 helpers	4 variants kept	success	13	$0.2598	pass: RTX A4500
t2-layernorm-forward layernorm_forward.cu	correct, 0 helpers	6 variants kept	budget	13	$0.4512	pass: RTX 2000 Ada
t2-softmax-forward softmax_forward.cu	correct, 45 helpers	8 variants kept	budget	14	$0.5402	not run

t2-classifier-fused classifier_fused.cu	correct, 0 helpers	5 variants kept	budget	14	$0.4562	pass: RTX A4500
t2-layernorm-backward layernorm_backward.cu	correct, 0 helpers	2 bodies for 9 variants	success	11	$0.7379	pass: RTX 2000 Ada
t3-matmul-backward matmul_backward.cu	correct, 0 helpers	1 variant kept	max_iterations	16	$0.2795	CPU only (oneMath)
t3-trimat-forward trimat_forward.cu	correct, 0 helpers	5 variants kept	success	9	$0.2112	CPU only (oneMath)
t3-matmul-backward-bias matmul_backward_bias.cu	correct, 0 helpers	8 variants kept	budget	14	$0.5007	pass: RTX 2000 Ada
t3-matmul-forward matmul_forward.cu	correct, 0 helpers	4 variants kept	success	15	$0.4046	CPU only (oneMath)

Muse ports llm.c's attention kernels: `attention_backward.cu` ported, `attention_forward.cu` cut off

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-attention.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t3-attention-backward attention_backward.cu	correct, 0 helpers	8 variants kept	max_iterations	20	$0.9233	pass: RTX 2000 Ada	-
t3-attention-forward attention_forward.cu	error	not checked	error	5	$0.2366	CPU only (oneMath)	-

Muse ports llm.c's `attention_forward.cu` (retry)

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-attention-forward.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t3-attention-forward attention_forward.cu	correct, 0 helpers	6 variants kept	max_iterations	20	$0.7639	pass: L4	-

matmul_backward re-ported with column-major oneMath calls

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-matmul-backward-colmajor.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t3-matmul-backward matmul_backward.cu	correct, 0 helpers	1 variant kept	success	7	$0.0943	pass: RTX 2000 Ada	correct, 0 helpers

Reruns with a stricter goal (plan C1-C3), Oct 27

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-reruns.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-fused-residual-forward fused_residual_forward.cu	error	not checked	error	6	$0.1874	not run	correct, 53 helpers
t2-softmax-forward softmax_forward.cu	verify fail	8 variants kept	max_iterations	14	$0.5534	not run	correct, 45 helpers
t2-layernorm-backward layernorm_backward.cu	error	not checked	error	10	$0.6563	not run	correct, 0 helpers

Reruns, second round (plan C2-C3), Oct 27

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-reruns-2.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-fused-residual-forward fused_residual_forward.cu	correct, 45 helpers	6 variants kept	budget	12	$0.5391	not run	correct, 53 helpers
t2-layernorm-backward layernorm_backward.cu	correct, 0 helpers	9 variants kept	budget	14	$0.9111	not run	correct, 0 helpers

`softmax_forward` again, with the stricter goal (Oct 27)

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-softmax-rerun.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t2-softmax-forward softmax_forward.cu	verify fail	8 variants kept	budget	19	$0.7470	not run	correct, 45 helpers

`softmax_forward` with more room: passes (Oct 27)

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-softmax-rerun-2.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t2-softmax-forward softmax_forward.cu	correct, 0 helpers	8 variants kept	success	23	$0.9063	pass: MI350X VF (vs reference) pass: L4	correct, 45 hel

Tier 1 without feedback (plan item E1, Oct 28)

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-baseline.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	model_finished	6	$0.0673	not run	correct,
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	model_finished	9	$0.0968	not run	correct,
t1-permute permute.cu	correct, 0 helpers	1 variant kept	model_finished	8	$0.0774	not run	correct,
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	model_finished	7	$0.0801	not run	correct,
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	model_finished	8	$0.0859	not run	correct,
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	max_iterations	12	$0.1488	not run	correct,
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	max_iterations	12	$0.1667	not run	correct,
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	model_finished	9	$0.0890	not run	correct,
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	model_finished	10	$0.1205	not run	correct,

t1-fused-residual-forward fused_residual_forward.cu	correct, 0 helpers	5 bodies for 6 variants	model_finished	10	$0.2478	not run	correct,

port-llmc-tier1-baseline-2

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-baseline-2.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	model_finished	9	$0.1087	not run	correct, 0 helpe
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	model_finished	8	$0.0738	not run	correct, 0 helpe
t1-permute permute.cu	correct, 0 helpers	1 variant kept	model_finished	8	$0.0699	not run	correct, 0 helpe
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	model_finished	7	$0.0958	not run	correct, 0 helpe
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	model_finished	7	$0.0905	not run	correct, 0 helpe
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	model_finished	9	$0.1658	not run	correct, 0 helpe
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	max_iterations	12	$0.1940	not run	correct, 0 helpe
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	model_finished	8	$0.0848	not run	correct, 0 helpe
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	model_finished	7	$0.0999	not run	correct, 0 helpe
t1-fused-residual-forward fused_residual_forward.cu	correct, 98 helpers	6 variants kept	max_iterations	12	$0.2779	not run	correct, 53 help

port-llmc-tier1-baseline-3

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-baseline-3.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	model_finished	7	$0.0649	not run	correct, 0 helpe
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	model_finished	11	$0.1066	not run	correct, 0 helpe
t1-permute permute.cu	correct, 0 helpers	1 variant kept	model_finished	7	$0.0581	not run	correct, 0 helpe
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	max_iterations	12	$0.1355	not run	correct, 0 helpe
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	model_finished	10	$0.1454	not run	correct, 0 helpe
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	model_finished	11	$0.0968	not run	correct, 0 helpe
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	max_iterations	12	$0.1171	not run	correct, 0 helpe
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	model_finished	8	$0.0790	not run	correct, 0 helpe

t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	model_finished	9	$0.0829	not run	correct, 0 helpe
t1-fused-residual-forward fused_residual_forward.cu	correct, 98 helpers	6 variants kept	max_iterations	12	$0.2441	not run	correct, 53 help

Tier 1 without SYCLomatic (Oct 27)

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-nomigrate.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	success	4	$0.0191	not run	correct, 0 helpers
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	success	4	$0.0234	not run	correct, 0 helpers
t1-permute permute.cu	correct, 0 helpers	1 variant kept	success	5	$0.0207	not run	correct, 0 helpers
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	success	5	$0.0284	not run	correct, 0 helpers
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	success	6	$0.0498	not run	correct, 0 helpers
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	success	4	$0.0382	not run	correct, 0 helpers
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	success	6	$0.0654	not run	correct, 0 helpers
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	success	8	$0.0689	not run	correct, 0 helpers
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	success	7	$0.0732	not run	correct, 0 helpers
t1-fused-residual-forward fused_residual_forward.cu	correct, 0 helpers	6 variants kept	success	4	$0.0888	not run	correct, 53 helpers

port-llmc-tier1-run3

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-run3.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	success	7	$0.0663	not run	correct, 0 helpe
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	success	6	$0.0655	not run	correct, 0 helpe
t1-permute permute.cu	correct, 0 helpers	1 variant kept	success	7	$0.0552	not run	correct, 0 helpe
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	success	6	$0.0615	not run	correct, 0 helpe
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	success	8	$0.0895	not run	correct, 0 helpe
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	success	6	$0.0779	not run	correct, 0 helpe
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	success	7	$0.0933	not run	correct, 0 helpe

t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	success	9	$0.1348	not run	correct, 0 helpe
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	success	8	$0.1038	not run	correct, 0 helpe
t1-fused-residual-forward fused_residual_forward.cu	correct, 0 helpers	5 bodies for 6 variants	success	7	$0.1623	not run	correct, 53 help

port-llmc-tier1-run4

Another run of these items, reported beside the others. What changed is in evidence/port-llmc-tier1-run4.

ITEM	HIDDEN GRADE	VARIANTS	LOOP	TURNS	COST	GPU	FIRST RUN
t1-crossentropy-forward crossentropy_forward.cu	correct, 0 helpers	1 variant kept	success	7	$0.0689	not run	correct, 0 helpe
t1-crossentropy-softmax-backward crossentropy_softmax_backward.cu	correct, 0 helpers	1 variant kept	success	6	$0.0630	not run	correct, 0 helpe
t1-permute permute.cu	correct, 0 helpers	1 variant kept	success	7	$0.0621	not run	correct, 0 helpe
t1-residual-forward residual_forward.cu	correct, 0 helpers	2 variants kept	success	7	$0.0574	not run	correct, 0 helpe
t1-gelu-forward gelu_forward.cu	correct, 0 helpers	2 variants kept	success	11	$0.1244	not run	correct, 0 helpe
t1-gelu-backward gelu_backward.cu	correct, 0 helpers	2 variants kept	success	9	$0.1118	not run	correct, 0 helpe
t1-encoder-forward encoder_forward.cu	correct, 0 helpers	3 variants kept	success	9	$0.1519	not run	correct, 0 helpe
t1-encoder-backward encoder_backward.cu	correct, 0 helpers	2 variants kept	success	6	$0.0579	not run	correct, 0 helpe
t1-adamw adamw.cu	correct, 0 helpers	2 variants kept	success	6	$0.0744	not run	correct, 0 helpe
t1-fused-residual-forward fused_residual_forward.cu	correct, 98 helpers	6 variants kept	max_iterations	12	$0.2925	not run	correct, 53 help

Muse spend

PORT-LLMC-ATTENTION $1.1599 muse-spark-1.3	PORT-LLMC-ATTENTION-FORWARD $0.7639 muse-spark-1.3	PORT-LLMC-MATMUL-BACKWARD-COLMAJOR $0.0943 muse-spark-1.3
PORT-LLMC-RERUNS $1.3971 muse-spark-1.3	PORT-LLMC-RERUNS-2 $1.4502 muse-spark-1.3	PORT-LLMC-SOFTMAX-RERUN $0.7470 muse-spark-1.3
PORT-LLMC-SOFTMAX-RERUN-2 $0.9063 muse-spark-1.3	PORT-LLMC-TIER1 $1.6698 muse-spark-1.3	PORT-LLMC-TIER1-BASELINE $1.1803 muse-spark-1.3
PORT-LLMC-TIER1-BASELINE-2 $1.2613 muse-spark-1.3	PORT-LLMC-TIER1-BASELINE-3 $1.1303 muse-spark-1.3	PORT-LLMC-TIER1-NOMIGRATE $0.4759 muse-spark-1.3
PORT-LLMC-TIER1-RUN2 $1.0064 muse-spark-1.3	PORT-LLMC-TIER1-RUN3 $0.9100 muse-spark-1.3	PORT-LLMC-TIER1-RUN4 $1.0642 muse-spark-1.3

PORT-LLMC-TIER2

$2.4454

muse-spark-1.3

PORT-LLMC-TIER23-RUN2

$2.7433

muse-spark-1.3

PORT-LLMC-TIER3

$1.3960

muse-spark-1.3

GPT-2 STAGES

$3.1064

3 loop runs

PLAN

$0.2648

one grounded plan

ALL OF THE ABOVE

$25.1728

ports, GPT-2 and plan

Built by `museport report` from the committed evidence in checocadena/museport. Every number on this page is read from a file under `evidence/`.